ALU HW2:
FLP Fused Multiply-Add (FMA)

Outlines

- fixed-point (FXP) fused-multiply-add (FMA) D=A*B+C
  - three synthesis constraints (area-opt, delay-opt, in-between)
- non-pipelined floating-point (FLP) FMA
  - alignment of C at the same time when A*B
  - named FLP_FMA
- pipelined FLP_FMA_4
  - 4 pipelined stages, named FLP_FMA_4
    - ✓ unpack, FXP multiply, align C,
    - ✓ FXP add
    - ✓ normalize
    - ✓ round, pack
- placement-and-routing (P&R) of FLP_FMA_4
  - mark the four stages in the layout view

FMA (FLP_FMA)

- combine the multiply and add operation: D=A*B+C
  - inputs A, B, C, and output D are all 32-bit IEEE formats
  - (sign, exponent, mantissa) for A, B, P=A*B, C
    - ✓ denoted as (sa, ea, ma), (sb, eb, mb), (sp, ep, mp), (sc, ec, mc)
  - alignment shift for C at the same time when computing the product mp=ma*mb
    - ✓ reduce FMA latency (cp. with alignment shift mp or mc), why?
    - ✓ might **left or right** alignment shift mantissa of C depending on the exponent difference of ep=ea+eb and ec
      - e.g. if ep-ec = -w, alignment left-shift mc by w bits
    - ✓ could consider only -26 < ep – ec < 48, why?
- add a control unit to consider special numbers (zero, infinity, NaN) and sub-normal for inputs
  - e.g., D=C if A=0 or B=0
  - also consider overflow/underflow for the output D
    - ✓ e.g., when ea+eb > 127 or < -149

alignment only for -26 <= ep – ec <= 48

- (sign, exponent, signed mantissa)
  - P=A*B: (sp, ep=ea+eb, mp=ma*mb), C: (sc, ec, mc)
  - |mc| < 2 (-2 < mc <=1, or 1 <= mc <2), i.e., mc : Q2.23
  - |mp| < 4 (-4 < mp <=1, or 1 <= mp <4), i.e., mp: Q3.46
- when ep > ec, and ep – ec >= 49
  - alignment-right-shifted mc -> mc'
  - |mc'| < 2^{-48} (because |mc| < 2)
  - |mc'| impact < 2^{-47} due to possible 1-bit left shift of normalization
    - ✓ mc' does NOT affect add result after rounding mp+mc' to ulp = 2^{-23}
- when ec > ep, and ec – ep >=27
  - alignment-right-shifted mp -> mp'
  - |mp'| < 2^{-25} (because |mb| < 4)
  - impact on | mc+mp' | < 2^{-24} due to possible 1-bit left shift of normalization
    - ✓ mp' could be ignored after rounding mc+mp' to ulp = 2^{-23}
  - *right*-shift mp by ec-ep bits to mp' and mc+mp' is equivalent to *left*-shift mc by ec-ep bits to get mc' and then mc'+mp
- **Q28.48** for converting mp, mc' to signed
  - 2 extra LSB round bits for mp_signed (Q3.46) => 48 fractional bits in total
  - 26 extra MSB bits for mc'_signed (Q2.23) => 28 integer bits

Pipelined FMA (FLP_FMA_4)

- 1st stage
  - unpack
  - ep=ea+eb
  - alignment shift of mc -> mc’
  - convert mc’ to signed
  - unsigned multiply mp= ma * mb
  - convert mp to signed
- 2nd stage
  - FXP add
  - convert signed result to sign-magnitude
- 3rd stage
  - LOD, normalization shift
- 4th stage
  - round
  - pack

1st stage

- similar to FLP_adder_pipe_4 in HW1 except
  - ep = ea+eb
  - e_diff = ep - ec
  - align signed mc at same time performing ma*mb
  - aligment of mc might be right-shift or left-shift
    - ✓ depending on e_diff = ep – ec is positive or negative
  - only consider -26 <= e_diff <= 48

unsigned multiplication and then conversion to signed

- a: (sa, eb, ma), b: (sb, eb, mb), c: (sc, ec, mc)
- unsigned multiplication ma*mb
  - p = ma * mb with unsigned input Q1.23
  - unsigned p < 4 in Q2.46
  - latch-gating for a, b fin case of a=0 or b=0
  - clock-gating for product in the pipelined registers
- alignment of mc and conversion to signed
  - left- or right-shift mc and d -> mc' in Q28.48
    - ✓ fractional bits include 46 + 1 (rounding bit)
    - ✓ integer bits include 1 (signed) + 24 (mc) + 2 (p<4)
- conversion of unsigned p Q2.48 p to signed Q28.48
  - signed_p = (sa*sb == 1) ? $signed(p) : $signed(-p);
- signed addition of signed_p + mc'

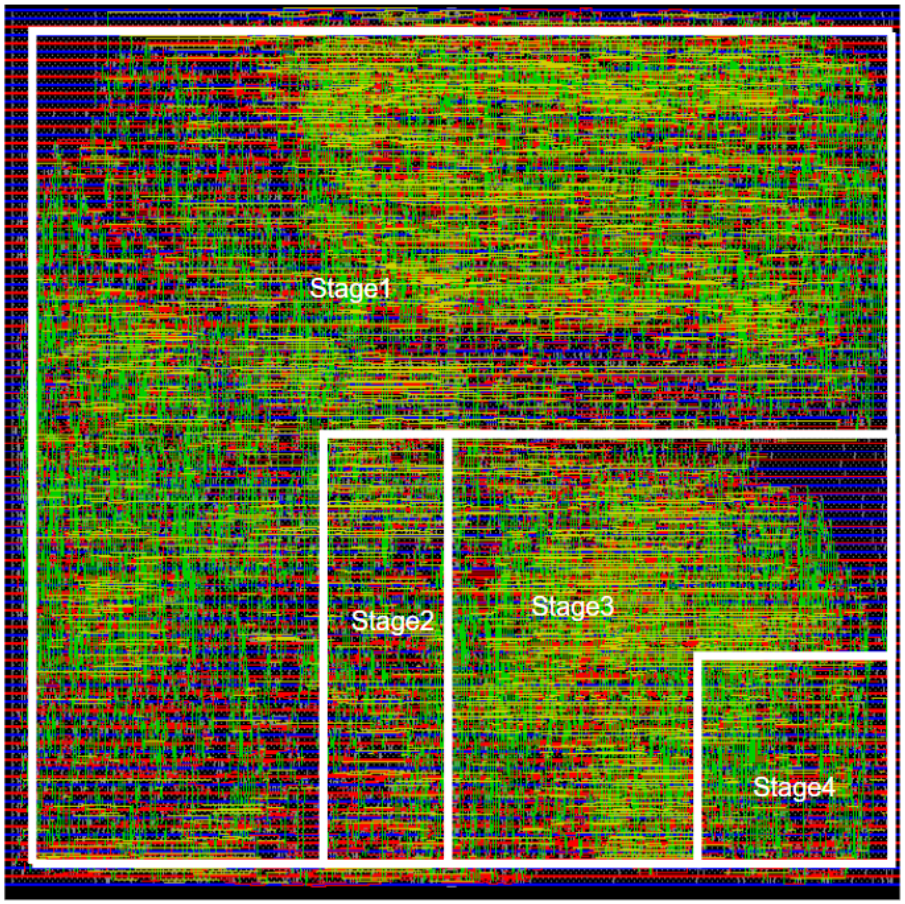
special condition

- add control unit to consider special situations
- overflow or under flow
- clock gating if some operands are zero
  - output D = A*B+C = C if A=0 or B=0
    - ✓ skip multiplication and addition
  - output D = A*B if C = 0
    - ✓ skip addition
- consider special number NaN
  - output D = Nan if any one of A, B, C is NaN

comparison table

- table of summary

		Area (um²)			Delay	Latency	Power (W)		
		CL	SL	Total	(ns)	(ns)	dynamic	leakage	total
FXP_FMA	area								
	mid								
	delay								
FLP_FMA	area								
	mid								
	delay								
FLP_FMA_4	area								
	mid								
	delay								

P&R of FLP_FMA_4

- perform automatic placement and roluting for the design FLP_FMA_4
- mark the four pipelined stages in the layout view